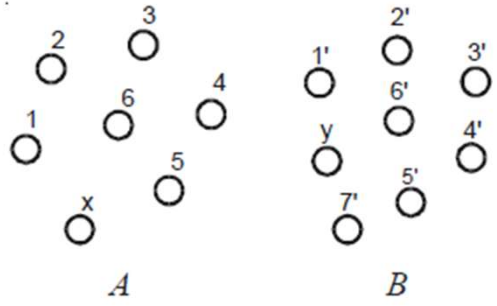
Inductance of Composite Conductor Lines—Self and Mutual GMDs

Consider a single phase line consisting of two parallel conductors A and B, conductor A consisting of x and conductor B of y strands, as illustrated in Fig. 5.7.

Let the conductors A and B carry currents I and –I respectively (since conductors of a 2-wire line carry the same current but in opposite directions).



Assuming uniform current density in both the conductors the current carried by each strand of conductor A will be $\frac{I}{x}$ while that carried by each strand of conductor B will be $\frac{-I}{y}$

Using Eq. (5.12) the flux linkages of strand 1 in conductor A can be written as

$$\psi_1 = 2 \times 10^{-7} \frac{I}{x}\left[\log_e \frac{1}{r_1'} + \log_e \frac{1}{d_{12}} + \log_e \frac{1}{d_{13}} + \ldots + \log_e \frac{1}{d_{1x}}\right]$$

$$- 2 \times 10^{-7} \frac{I}{y}\left[\log_e \frac{1}{d11'} + \log_e \frac{1}{d12'} + \log_e \frac{1}{d13'} + \ldots + \log_e \frac{1}{d_{1y}}\right]$$

$$\text{or } \psi_1 = 2 \times 10^{-7} \, I \log_e \frac{\sqrt[y]{(d_{11'} \; d_{12'} \; d_{13'} \ldots d_{1y})}}{\sqrt[x]{r_1' \, d_{12} \; d_{13} \ldots d_{1x}}} \text{ Wb-turns/m}$$

Inductance of strand 1 of conductor A

$$L_1 = \frac{\psi_1}{\frac{I}{x}} = 2 \times 10^{-7} \, x \log_e \frac{\sqrt[y]{(d_{11'} \; d_{12'} \; d_{13'} \; d_{14'} \ldots d_{1y})}}{\sqrt[x]{r' \, d_{12} \; d_{13} \; d_{14} \ldots d_{1x}}} \quad \ldots(5.13)$$